

HollowHaven

A Context-Aware Artificial Tree Hollow for Greater Gliders

Project Report

From Ecological Problem to Prototype Evaluation

COMP5047

The University of Sydney

November 2026

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ABSTRACT

HollowHaven investigates how an artificial tree hollow can support habitat care for the southern greater glider through environmental sensing, controlled water delivery and low-disturbance observation. The project developed from a broad adaptive-shelter proposal into an ESP32 prototype with a temperature-and-light-gated valve, dual-point wetting checks, maintenance warnings and event-triggered audio. This report connects the ecological problem to the design decisions, implementation and available evaluation evidence. Five host-side software test executables passed during report preparation. Separately, the team provided and confirmed ten days of measured microclimate records. Within its daily observation window, the drip condition was on average 0.51 °C cooler than the artificial hollow without dripping, with a maximum simultaneous difference of 2.20 °C. These are descriptive measurement results; they do not by themselves establish animal-welfare benefits. The principal outcome is an integrated, testable habitat-care prototype and a clearer programme for physical validation.

1. DISCOVERING THE PROBLEM: HABITAT LOSS AND SHELTER QUALITY

1.1. Australian ecological context and non-human actors

The focal actor is the southern greater glider (*Petauroides volans*), which the NSW Threatened Species Scientific Committee listed as Endangered in 2022 [1]. Greater gliders depend on mature trees and suitable hollows for shelter. Wildlife Queensland describes habitat enhancement, nest-box installation where hollow-bearing trees are limited, and population monitoring as conservation activities [2]. These establish an identifiable Australian setting for the project: hollow-dependent animals living in forest landscapes where suitable shelter cannot simply be assumed to remain available.

Ridley et al. identify the loss of hollow-bearing trees as a driver of decline for hollow-dependent species, with wildfire exacerbating the shortage. Their southern greater glider study installed nest boxes at sites in New South Wales and Victoria and compared these with control sites. Its early results did not show a significant difference in relative abundance between nest-box and control sites [3]. It places supplementary shelter within a tested conservation response, rather than treating a nest box as proof of population recovery.

The original project proposal identified clearing, logging and severe bushfire as pressures on hollow-bearing habitat and selected a greater-glider-focused shelter as its response [4]. That proposal is evidence of the team's motivation, not a substitute for ecological literature. The key design implication is that providing a cavity and providing a suitable microclimate are distinct tasks. An artificial hollow is supplementary infrastructure; it cannot reproduce the wider ecological functions of an intact mature forest or justify removing natural hollow-bearing trees.

Other animals may encounter the same cavity, including non-target hollow users and insects. The supporting tree is also affected by mounting and maintenance. Human stakeholders include carers, researchers and maintainers. This wider set of encounters matters because a sensor trigger cannot identify an animal's species, establish its comfort or determine that intervention is warranted.

1.2. Established artificial shelters and their limits

Greening Australia's official account of its collaboration with the Australian National University and WWF-Australia describes nest boxes incorporating insulation, air gaps and heat-reflective, fire-resistant, non-toxic coatings. Cameras documented greater-glider use, while the project emphasised that artificial shelters supplement rather than replace hollow-bearing forest protection [5]. This provides a concrete comparator: passive thermal design and ongoing observation are

already important conservation practices. HollowHaven adds inspectable watering and maintenance functions; it does not claim to outperform that project.

The Australian National University's account of the same collaboration adds the design and testing context: traditional thin-walled boxes can become much hotter than tree hollows; the improved boxes underwent laboratory thermal tests, with box sites planned for comparison against control sites without boxes [6]. This is an additional account of the same project, not an independent second experiment.

Brown et al. directly studied southern greater glider use of nest boxes. Over two years, 11 of 34 installed boxes were used by the species, showing that boxes can provide supplementary shelter. The study also found that the boxes provided less thermal insulation than tree hollows [7]. Occupancy therefore demonstrates use, but does not alone establish thermal suitability or population benefit.

1.3. Thermal evidence and the HollowHaven design hypothesis

Griffiths et al. compared natural cavities with plywood nest boxes, log hollows and chainsaw-carved cavities, using temperature records over 110 days. Carved cavities in living trees more closely reproduced natural-hollow thermal profiles; tested nest boxes were substantially hotter during the day, while log hollows had intermediate behaviour [8]. The study concerns its sampled shelters and animals, rather than demonstrating performance for our greater-glider prototype.

Rowland et al. measured paired nest-box and tree-hollow microclimates for arboreal marsupials. Their boxes were more influenced by ambient temperature and solar radiation, with summer maximum temperatures averaging about 8 °C above those in the sampled hollows [9]. In a New South Wales field comparison of boxes for sugar and squirrel gliders, Honey et al. found that roof insulation reduced maximum temperatures by an average of 3.1 °C, although boxes could still reach high temperatures during heatwaves [10]. These studies justify evaluating thermal performance; their temperature differences are not measurements of southern greater gliders or HollowHaven.

Table 1. Thermal design lessons from the comparative study of Griffiths et al. (2018). These are contextual findings, not HollowHaven test results.

Shelter	Distinction	Design implication
Natural hollow	Reference cavity within a tree.	A cavity's surrounding structure matters alongside its empty volume.
Plywood nest box	Stronger daytime heating in the study.	An available nesting space need not provide equivalent thermal buffering.
Log hollow	Less variable than boxes but not equivalent to natural hollows.	Natural-looking material alone does not establish thermal equivalence.
Chainsaw-carved cavity	Closer correspondence to natural thermal profiles.	Passive structure is a serious comparator for any active cooling proposal.

We therefore treat enclosure construction, siting and active control as related design questions. Bark-like appearance cannot stand in for insulation measurements. Likewise, adding a valve cannot establish a benefit without comparing otherwise similar shelters under comparable environmental conditions. These are our design inferences from the literature. We do not claim that all nest boxes overheat or assign an unverified species-specific safe temperature.

The Australian Government's Your Home passive-cooling guidance explains that evaporation absorbs heat and that its effectiveness depends on humidity, airflow and exposed water area. It also identifies water consumption as a practical cost [11]. This supports the physical rationale for the guiding layer and the decision to monitor humidity alongside temperature. The guidance concerns buildings, not animal-specific safety thresholds. Our thresholds and fault rules are engineering

choices requiring ecological validation.

On that basis, HollowHaven proposes a testable design hypothesis: directing water to an exterior guiding layer, with monitored temperature and humidity and controlled delivery, may moderate an artificial hollow's microclimate while making delivery and maintenance inspectable. The evaporation guidance and preceding shelter studies do not establish that this particular device is effective or safe for southern greater gliders. Physical controlled comparisons and animal-level evaluation remain necessary.

2. DEFINING THE DESIGN PROBLEM

The problem was framed as: *How can an artificial hollow observe changing conditions and provide a limited, inspectable response, while preserving animal choice and keeping maintenance accountable?* This formulation links a shelter's physical conditions with the practical difficulty of knowing whether a response actually occurred.

Four requirements followed. First, observe temperature and humidity rather than assuming the exterior appearance represents internal conditions. Second, provide a controllable water-delivery mechanism and distinguish commanding it from confirming wetting. Third, detect maintenance needs without interpreting every anomaly as a biological emergency. Fourth, record possible visits with minimal demands on the animal and retain uncertainty when interpreting them.

Success was separated into three levels: correct software behaviour, physically effective delivery and thermal response, and eventual ecological suitability. The first can be tested with controlled inputs; the second needs instrumented hardware experiments; the third requires appropriate field observation and expertise. This separation shaped both the prototype and the evaluation reported below.

2.1. Animal–Computer Interaction as a design framework

Mancini's Animal–Computer Interaction (ACI) manifesto calls for animals' perspectives to inform technologies that affect them [12]. We apply that position through concrete choices. An occupant should not need to learn a command or perform a task to obtain shelter. It should be able to enter, leave or ignore the cavity. Presence is a useful observation but is not consent to all subsequent actuation, nor proof of welfare.

Our sensing-first approach avoids requiring acoustic lures, playback or animal vocal responses. The microphone listens; optical time-of-flight sensing detects nearby objects without acoustic ranging. To avoid unnecessary disturbance during daytime rest and preserve the existing sound environment as far as possible, the team did not adopt a continuously running fan and limited motor movement to changes in the water-valve state. Separate opening and closing temperature thresholds are also intended to reduce repeated switching near a boundary. Nevertheless, the servo and water flow may produce sound, and neither operating sound levels nor animal responses have been measured, so reduced disturbance remains an intention to test. The practical ethical aim is proportionate, reversible assistance. Installing and watering a shelter are interventions; responsible care requires evaluating their consequences rather than claiming an absence of interference. These principles are our application of ACI, not evidence that the current enclosure has passed an animal-welfare assessment.

3. DESIGN DEVELOPMENT: FROM PROPOSAL TO A BUILDABLE SYSTEM

3.1. Scope refinement

The early repository proposal explored ventilation fins, a fan, external ambient feedback and possible AI-assisted event classification [4]. The subsequent implementation centres on a servo-driven water path, environmental sensing and a local monitoring service. A continuously running fan was not carried into the current design, partly because the team wanted to avoid additional mechanical sound during animal rest; the present servo moves only when the water-valve state needs to change. The progression is documented by the repository's proposal and implementation artefacts, rather than reconstructed as an invented week-by-week development diary.

A light-controlled valve stage appears in commit `871d7ae`. The inspected snapshot `cad07a9` contains a temperature-and-light controller, dual moisture channels and associated tests [13]. The resulting scope allows a particular chain to be examined: environmental input, control decision, commanded movement, wetting evidence and maintenance indication. Earlier ideas such as autonomous deterrence or AI-selected animal-management modes are not presented as completed features.

3.2. Water delivery and observability

Water is directed towards a water-guiding layer as the basis of an evaporative-cooling concept. The design hypothesis is that controlled wetting can moderate temperature without unacceptable humidity or disturbance. This remains a hypothesis until physical tests establish heat transfer, actual water use and moisture reaching the resting space.

Two moisture points help distinguish a valve command from a successful delivery outcome. If the commanded fully open state has persisted for at least 60 seconds and either point remains dry, a possible blockage warning is produced. A separate low-water warning supports refilling. This makes failure visible, although dry material can have multiple causes; the warning is not a definitive diagnosis of blockage.

3.3. Enclosure and intended setting

The team's enclosure design uses a bark covering and similar exterior colours to evoke a tree hollow. The covering is intended to mimic the surrounding woodland and help the device blend visually into its setting, reducing its prominence as a human-facing object and reflecting the team's aim to limit its additional environmental footprint. No thermal conductivity, weatherproofness or animal preference is inferred from its appearance. Whether the covering has a lower environmental burden also depends on its material source, durability, effects on the supporting tree and maintenance requirements, which have not yet been assessed.

The intended animal encounter is an unobstructed cavity rather than an interactive screen. Mounting should protect the supporting tree, maintenance access should minimise disturbance, and wiring and moving parts should be separated from the accessible resting space. These are requirements for a field-ready enclosure, not verified specifications inferred from the code. Photographic fabrication records and measured enclosure dimensions are not part of the evidence analysed here.

The present prototype depends on local Wi-Fi, a nearby computer, external power and a refillable reservoir. These constraints place the demonstrated architecture closer to a supervised prototype than an unattended forest installation. Its presence communicates an ongoing responsibility to maintain artificial habitat, rather than a technological replacement for conservation of natural habitat.

3.4. Contribution relative to related work

The contribution lies in combining a shelter with a measurable control-and-maintenance process: local sensing, actuation, two-point wetting checks, refill warnings and event records. Compared with passive cavity approaches in the literature, this adds diagnostic visibility and active-response capability, but also power demand, moving parts, water dependence and maintenance burden. We do not claim global novelty or superior thermal performance over carved naturalistic cavities. The relevant question is whether the additional complexity produces a demonstrable net benefit.

4. COMPLETED PROTOTYPE AND CURRENT BEHAVIOUR

4.1. Integrated architecture

An ESP32 collects SHT3x temperature and humidity readings, two capacitive moisture channels, light and water-level signals, VL53L0X distance readings and SEN0487 microphone input. An MG90S servo operates the water valve. The computer receives telemetry and event audio, stores CSV and WAV records, and serves a browser interface. This distribution embeds sensing and actuation in the shelter while retaining human-readable records outside it [14, 15].

Installation steps, pin assignments, network configuration and routine operation are documented in the technical and user manuals [15, 16]. They are intentionally not repeated here. The report focuses on how the implementation addresses the design problem and what evidence supports its outcomes.

4.2. Current control logic

The inspected firmware opens the valve when the light input qualifies (valid ADC below 500) and valid temperature is strictly above 30 °C. After the servo reaches the commanded fully open position, the controller holds for at least 60 seconds and then checks temperature at one-minute intervals. It closes only when a valid reading is strictly below 26 °C. Subsequent light changes do not terminate the cycle; invalid temperature readings cause another interval of holding [14, 13].

The servo changes by 10 μs every 100 ms, taking approximately 13 seconds across the configured range. Reported valve position reflects the output command, not a mechanical position sensor. Low water currently raises a warning but does not inhibit opening. The 30/26 °C settings are engineering thresholds in this prototype; the cited ecological sources do not establish them as species-specific safety limits. This supersedes the earlier report's description of light-only control.

4.3. Visitor events and optional AI interpretation

Three consecutive valid distance readings at or below 189 mm trigger suspected occupancy; five at or above 220 mm clear it. This hysteresis reduces unstable transitions. Entry triggers approximately 30 seconds of audio, which is transferred and stored locally. A nearby object is not necessarily an animal, and neither the distance event nor the recording establishes species identity.

The operator can manually submit a recording to Gemini audio understanding or a configured classification service. Results remain candidate interpretations for human review. They do not change valve logic or confirm occupancy identity. Audio files are not automatically submitted to an external AI service. The present microphone and sampling arrangement also does not support comprehensive ultrasonic bat identification [15, 16].

5. PROTOTYPE TESTING AND EVALUATION METHOD

5.1. Data provenance

The analysis uses repository snapshot `cad07a9`. The team has confirmed that the ten-day comparison series are actual measurements and that the old SIMULATED labels and modelling descriptions are legacy documentation errors. This report follows that correction without changing numerical values. Original exports are retained for traceability. Calendar dates, site details, individual instruments, calibration records and experimental layout were not supplied in full. We therefore report descriptive comparisons without inventing these conditions or claiming that all confounding factors have been controlled.

Software tests evaluate implementation rules on a host computer. The measured records evaluate temperature, humidity and operating state. Neither alone establishes animal-welfare improvement.

5.2. Host-side functional tests

Five existing C test executables were built with MSVC 19.39 through CMake and run using CTest in Debug configuration, retaining assertions. The run took place during preparation of this revision, not as a claimed historical team test. The tests cover valve control, maintenance warnings, moisture classification, presence detection and telemetry. Hardware drivers, wireless performance, servo mechanics and acoustic classification were outside this test run.

5.3. Measurement records and analysis

The dataset covers ten days at five-minute intervals: 2,880 full-day records and 960 records within the daily 09:00–16:55 observation window. Days 4 and 8 are labelled cloudy; the others are sunny. Series include ambient conditions, a standard box, a natural hollow and artificial hollows with and without dripping [17].

We checked row counts, unique day/time pairs, complete key numeric fields and exact agreement of the observation-window export with the full-day subset. Paired temperature difference is no-drip minus drip at the same timestamp. Daily peak reduction compares each series' separate maximum, which need not occur simultaneously. Threshold exposure and opening duration are sample-count estimates multiplied by five minutes. The 30 °C threshold is an engineering setting, not an established biological safety limit.

6. RESULTS AND FINAL PROJECT OUTCOME

6.1. Functional results

All five host-side test executables passed (5/5; zero failures). Valve tests check strict temperature boundaries, the light entry condition, full-open timing, invalid readings and re-entry. Maintenance tests include the 60-second boundary and wet, dry and unknown states. Presence tests cover confirmation counts, hysteresis, invalid readings and repeated occupancy. Moisture and telemetry tests check their respective state and record logic. These results establish that the tested software assertions hold in the host environment; they do not certify sensor accuracy or physical delivery.

6.2. Measured thermal response

Within the observation window, mean temperatures were 35.54 °C for the standard box, 28.20 °C for the natural hollow, 28.76 °C for the artificial hollow without dripping, and 28.25 °C with dripping. The paired mean reduction attributable to adding dripping within these records was 0.51 °C, with a maximum simultaneous reduction of 2.20 °C.

The much larger difference between the standard box and drip model combines enclosure differences with watering. It must not be presented as a measured cooling effect of the valve. Likewise, closeness to the hypothetical natural-hollow series is insufficient to establish equivalence in microclimate, use or ecological function.

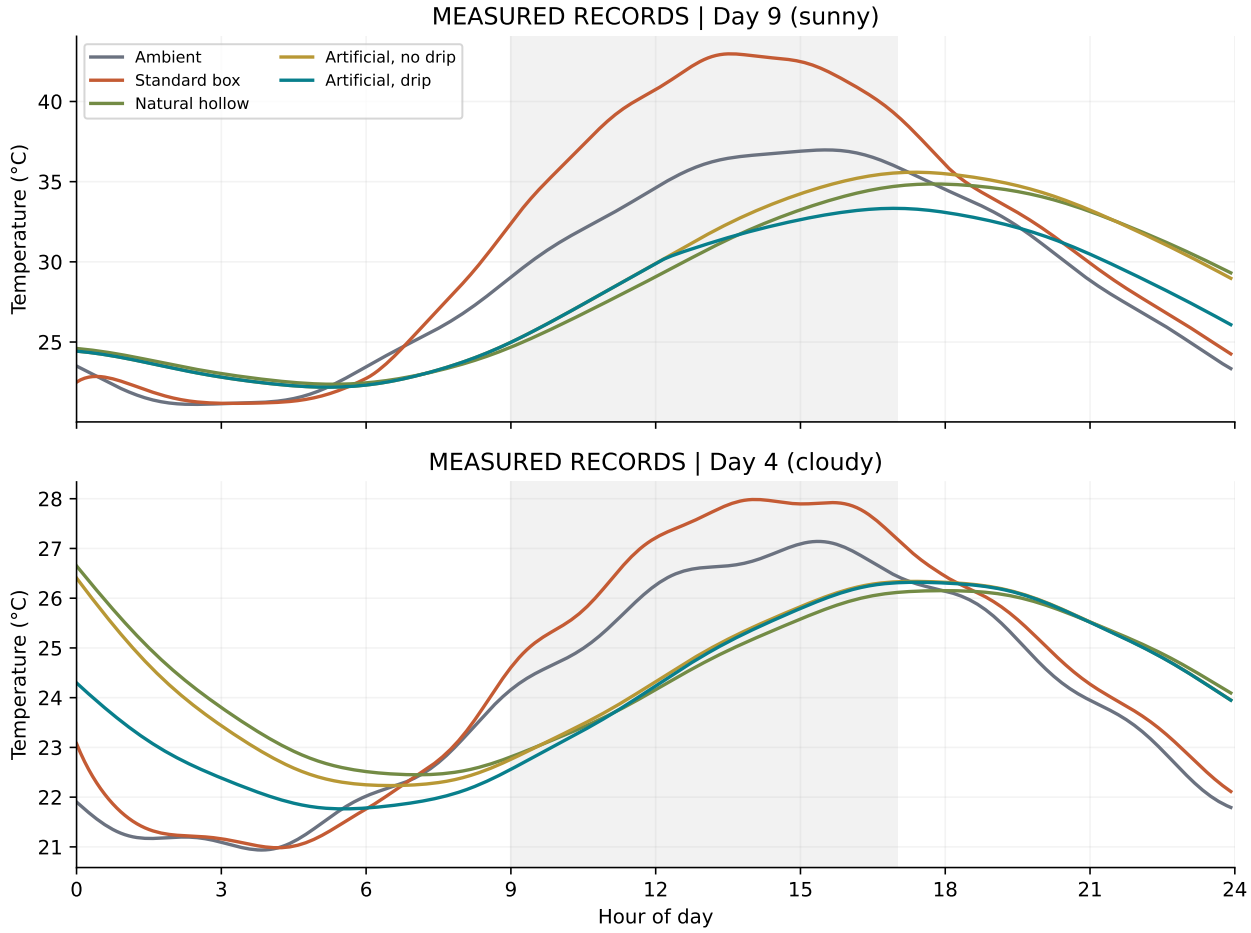


Fig. 1. Measured temperature profiles for Day 9 (the highest standard-box observation-window peak) and Day 4 (cloudy). Shading marks 09:00–17:00. Full-day curves expose delayed peaks and night-time behaviour that a daytime window can omit. Source: team-provided measurement CSV; plotted for this report.

6.3. Humidity, threshold exposure and operating cost

Mean observation-window relative humidity was 49.56% without dripping and 50.76% with dripping: an increase of 1.20 percentage points. The records do not separate changes due to cooling from changes in water-vapour content; excess humidity remains a design concern.

Across the complete 240 recorded hours, the no-drip and drip artificial-hollow conditions respectively accumulated approximately 72.33 and 50.33 sample-hours above 30 °C, a difference of 22.00 hours. This is an engineering threshold

Table 2. Measured daily peak temperatures within 09:00–16:55, in °C. Reduction is the no-drip peak minus the drip peak, not a simultaneous difference. Values are team-confirmed measurements.

Day	Weather	Box	Natural	No drip	Drip	Reduction
1	Sunny	39.50	32.20	32.87	31.00	1.87
2	Sunny	40.76	33.06	33.79	31.76	2.04
3	Sunny	37.12	30.56	31.15	30.13	1.02
4	Cloudy	27.99	26.11	26.33	26.31	0.02
5	Sunny	41.79	33.82	34.61	32.50	2.12
6	Sunny	40.05	32.62	33.31	31.35	1.95
7	Sunny	38.32	31.36	31.99	30.41	1.58
8	Cloudy	29.27	26.96	27.22	27.19	0.02
9	Sunny	42.97	34.70	35.53	33.34	2.20
10	Sunny	40.41	33.08	33.79	31.75	2.04

exposure metric, not a validated measure of physiological safety. The recorded valve was open for approximately 79.67 sample-hours, showing that the selected rules can require prolonged opening. Without measured flow rate, reservoir size and evaporation, this cannot be converted into water consumption or refill frequency.

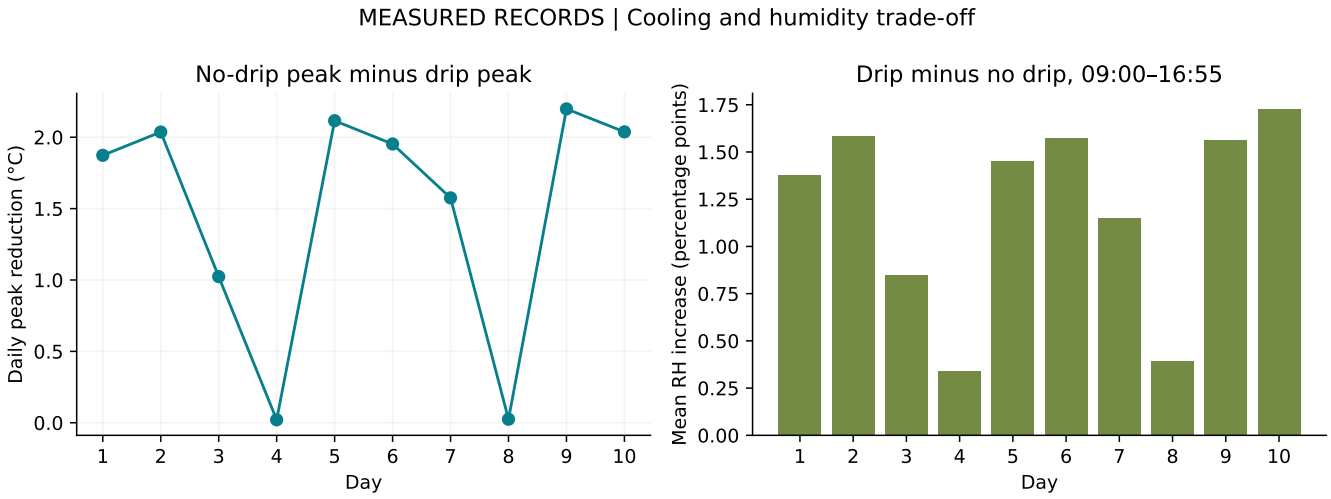


Fig. 2. Measured observation-window comparisons: daily peak temperature reduction and mean relative-humidity increase. Cooling and humidity must be assessed together. Source: repository CSV, recalculated for this report.

6.4. Design implications of the evaluation

The recorded drip condition was cooler in this comparison, but the mean difference, humidity change and opening time need to be interpreted together. Effective wetted area, airflow and ambient humidity may influence evaporation, but the present records cannot separate their effects. Table 3 translates the observations into questions for the next design iteration. These are inferences and proposed changes, not completed improvements or established explanations of the physical cause of the temperature difference.

Table 3. Observed results, design implications and proposed next steps; measurement and valve-state data from the ten-day records [17].

Observation	Design implication	Proposed next step
Paired mean difference of 0.51 °C; maximum simultaneous difference of 2.20 °C	The records show cooling, but the overall mean effect is modest, with larger differences at some times. The cause cannot be isolated from these data alone.	Compare the water-guiding layer's area, position and actual wetting extent, alongside passive insulation controls.
Sunny-day peak reductions of 1.02–2.20 °C; about 0.02 °C on each of two cloudy days, with no open-valve samples	The near-zero cloudy-day differences coincide with no recorded valve opening; they do not establish poor evaporation under clouds. Sunny-day differences may reflect temperature, light and humidity together.	Compare active and inactive control conditions while recording irradiance, humidity, wind, valve state and placement.
Mean relative humidity 1.20 percentage points higher with dripping	Cooling may accompany a change in internal humidity; the contributions of lower temperature and added water vapour remain unresolved.	Include humidity in feedback, investigate an ecologically validated candidate limit, and check for wetting of the resting space.
Valve open for about 79.67 sample-hours	The present rules may create substantial water and maintenance demand. Opening time alone does not measure water consumption.	Measure actual flow and reservoir capacity before evaluating a maximum opening time, low-water protection and revised control rules.
All five host-side software tests passed	The tested rules behaved as expected on the host computer; physical valve movement, delivery and sensor accuracy remain unverified.	Test the real valve, water flow, calibrated sensors and failure states on hardware.

Reproducible physical control tests should precede claims about long-term operation or animal benefit. Actual water use and operating sound should be measured alongside adverse outcomes such as excessive humidity or water reaching the resting space.

6.5. Practical outcome and next design iteration

HollowHaven now has a prototype architecture linking environmental sensing, valve control, wetting checks, maintenance warnings and event records, with optional human-reviewed event interpretation. All five host-side software tests passed, showing that the tested control rules behaved as expected. In the team-confirmed ten-day records, the drip condition averaged 0.51 °C below the no-drip condition within the observation window, with a maximum simultaneous difference of 2.20 °C and mean relative humidity 1.20 percentage points higher. The drip condition also accumulated less time above the engineering temperature threshold in the full-day records. These observations show cooling under the recorded conditions while exposing trade-offs among effect size, humidity and prolonged valve opening. The available tests do not yet establish actual water delivery by the physical valve, long-term reliability, field durability, greater-glider adoption or welfare benefit.

The next iteration should measure the physical device before revising its control strategy. We plan to measure actual flow, wetting extent, valve movement and operating sound, and to compare different guiding-layer areas and positions with passive insulation. Those measurements can inform a maximum opening time, low-water protection and the inclusion of humidity in control feedback; any humidity limit needs empirical evidence and ecological advice. Further matched trials should record light, wind, ambient humidity and valve state so that weather effects can be distinguished from whether the controller activated. Only after the physical effects, durability and possible adverse outcomes are assessed should field suitability and greater-glider use or avoidance be evaluated as evidence of longer-term ecological value.

7. LIMITATIONS AND CRITICAL REFLECTION

7.1. Experimental and ecological limitations

The ten-day records support descriptive analysis, but complete dates, site information, calibration and sensor placement remain undocumented in the supplied material. Adjacent observations are not independent experimental replicates. Timestamp matching does not eliminate differences in shade, ventilation, construction or sensor bias. The results cannot establish a universal cooling effect or be translated directly into animal welfare. No animal behaviour or occupancy-success measurements are available.

7.2. Technical limitations

The system lacks mechanical valve-position and direct flow feedback. ADC-based moisture classes are not calibrated water content, and the water-level warning is not a continuous volume measurement. Low-water readings do not inhibit opening. Invalid temperature during an open cycle currently keeps the valve open until a later check, potentially extending water use. Temperature participates in feedback, but humidity does not constrain actuation. These gaps are significant even if software tests pass, because the tests verify the selected rules rather than proving that the rules are ecologically appropriate.

7.3. Design trade-offs

Active watering introduces dependencies that passive shelters avoid. A bark-coloured enclosure may fit its setting visually while still performing poorly thermally. Remote observation may reduce the need to open a shelter, but it can also encourage overconfidence in an incomplete dashboard. An ACI-informed assessment must therefore include avoidance, disturbance and unwanted encounters, not simply the amount of data collected or the number of visits detected.

8. FUTURE DEVELOPMENT

The next priority is a matched physical comparison of the same enclosure with and without watering. Calibrated temperature and humidity probes should record inside and ambient conditions simultaneously, alongside actual water flow, wetting extent and operational sound. Repeated comparable trials should test solar exposure and airflow while recording adverse outcomes such as excess humidity or wetting of the resting area. Passive improvements should be evaluated before assuming active control is necessary.

The engineering priority is to add flow or valve feedback, low-water protection, bounded opening time and explicit responses to invalid sensors. Thresholds and humidity limits should then be revised with ecological advice and empirical evidence. Field preparation additionally requires enclosure documentation, mounting assessment, weatherproofing, power budgeting and a practical refill schedule. Only after these stages should appropriate, permitted wildlife observation assess use, avoidance and disturbance. Multi-unit monitoring is a later extension, not a substitute for validating one unit.

9. AI USE, VERIFICATION AND AUTHORSHIP

9.1. AI as part of the prototype

Optional audio interpretation is a bounded product feature: the human operator selects recordings, receives uncertain candidate labels and retains responsibility for interpretation. The control system does not defer valve decisions to a language model. No classification accuracy, species-confirmation rate or benchmark result is claimed without an appropriate labelled test set.

9.2. AI during research and report production

The team used ChatGPT, developed by OpenAI, to support brainstorming, outlining, translation between Chinese and English, literature discovery, language refinement, LaTeX formatting and revision of the report [18]. During this revision, AI also assisted with reading repository documentation, preparing data-analysis code and drafting explanations. These uses are disclosed separately from runtime audio analysis.

The team's policy is not to accept AI-provided information uncritically. Suggested sources must be traced to the actual publication or issuing organisation, technical descriptions checked against the implementation, and numerical claims recomputed from identified data. Unverifiable assertions should be removed or qualified. AI output is not treated as scientific evidence; the bibliography cites the underlying sources. The correction of legacy data labels and the changed temperature-control logic illustrate why cross-checking matters.

This revision recomputed the reported CSV metrics and reran the five software tests. The team retains responsibility for final source review, authorship, interpretation and any subsequent experimental claims. This disclosure describes the verification performed and policy adopted, rather than asserting a perfect audit of every prior AI interaction.

10. CONCLUSION

HollowHaven developed from an ecological shelter problem into an integrated prototype for observable, limited habitat intervention. Its present contribution is a temperature-and-light-driven water controller with wetting checks, maintenance visibility and event records. Host-side tests passed, while the measurement comparison showed modest cooling alongside humidity changes and prolonged opening. The remaining challenge is to show that the physical device delivers a net benefit under realistic conditions. Protecting natural habitat, preserving animal choice and testing adverse effects remain central to that next stage.

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